

Syllabus

§1 Day 1

1. Naive Set Theory: disjoint union
2. Equivalence relations
3. Canonical decomposition of a map
4. Definition of category
5. Isomorphisms
6. Initial and terminal objects
7. A commutative diagram
8. No HW.

§2 Day 2

1. What is a group?
2. Examples of groups
3. No HW.

§3 Day 3

1. Order of an element
2. Examples of order of elements
3. Property: if $g^n = e$, then order of g divides n .
4. symmetric groups (automorphism of set of 1 to n , $n!$, S_3)
5. dihedral groups D_{2n} (group element is operation)
6. cyclic groups ($\mathbb{Z}/n\mathbb{Z}$, who is the generator, order n)
7. units group $(\mathbb{Z}/n\mathbb{Z})^\times$, is there a generator? that is, is it cyclic? (this is very hard)
- 8.
9. HW: reading chapter 2 sections 1 and 2. Do 1.3, 1.4, 1.5, 1.8, 1.10, 1.11. Do 2.2, 2.5, 2.6, 2.7, 2.11, 2.12, 2.13, 2.15, 2.17

§4 Day 4

1. Group homomorphisms. $\phi(a * b) = \phi(a) \times \phi(b)$. Show $\phi(e) = e'$, $\phi(a^{-1}) = \phi(a)^{-1}$
2. Example: e as initial and final. $\mathbb{Z}$ to $\mathbb{Z}/n\mathbb{Z}$, $\mathbb{Z}$ to G by g^n . $(\mathbb{R}, +)$ to $(\mathbb{R}^\times, \times)$, S_3 to D_6 , S_3 to C_2 . $\det(M)$ as homomorphism from $GL_n(\mathbb{R})$ to $\mathbb{R}^\times$. $\mathbb{C}^\times$ to $\mathbb{R}^\times$ by $z \mapsto |z|$.
3. Category of groups.
4. Category of abelian groups.
5. Initial and final objects in groups: trivial group
6. Product groups: product is a universal object in some sense. What is product in sets? What is product in general. Product and coproduct in groups and abelian groups.
7. Some homomorphisms from and to product groups.
8. Group homomorphisms and order of elements.
9. Isomorphisms in general
10. groups: bijective homomorphisms (nontrivial, deserve a proof): $S_3 \cong D_6$, $\mathbb{Z}/6\mathbb{Z} \not\cong D_6$, $(\mathbb{Z}/7\mathbb{Z})^\times \cong \mathbb{Z}/6\mathbb{Z}$, $C_2 \times C_3 \cong C_6$.
11. homeomorphisms of topological spaces is not bijective morphism.
12. When not isomorphic? Key property, isomorphism preserves order of an element. Look at order of elements.
13. homomorphisms of abelian groups form a group, in fact, a ring.
14. automorphism of groups form a group.
15. HW: reading chapter 2, section 3, 4. Do 3.2, 3.4. Do 4.1, 4.3, 4.7, 4.8, 4.9, 4.10, 4.13, 4.14, 4.15, 4.17.

§5 Day 5

1. Universal properties in category theory: Given a category, a natural question is whether there are initial and final (not always exists). We have seen that initial and final objects are unique up to isomorphisms which is very nice. Initial and final objects are "extreme" objects in the category without posing any extra conditions. We can also ask for "extreme" objects with some extra conditions, and this is what universal properties are about: the way to do this is to construct a new category, very much similar to the old one but posing the condition we need. And we find the initial object in this new category. For example, we want to find an "extreme" object P with morphisms to A and B (A and B are fixed) We can ask for the "extreme" object with some extra conditions, for example, we want the morphism from P to A and B to be homomorphisms. This is what we call product of A and B .

2. Products in groups, sets, abelian groups and topological spaces.
3. Coproducts in sets, abelian groups and topological spaces.
4. HW: reading (chapter 1, section 5). Do all exercises from ch1 section 5.

§6 Day 6

1. Products in topological spaces.
2. Coproducts in sets, abelian groups and topological spaces.
3. Free groups and free abelian groups.
4. Universal property of free groups, free groups is universal so unique up to isomorphisms.
5. Construction of free groups (maybe skip)
6. Construction of free abelian groups.
7. Chapter 2 section 5 Do 5.1, 5.2, 5.3, 5.4, 5.5, 5.9, 5.10 in chapter 2 section 5.
Exercise: find the free objects in the category of topological spaces.

§7 Day 7

1. Definition of subgroups in terms of category theory: a subgroup of G is a group H with an injective homomorphism to G .
2. Standard definition of subgroup: a subset of G closed under group operation and inverse.
3. H is a subgroup of G if and only if for any $a, b \in H$, $ab^{-1} \in H$. The most useful criterion for checking whether a subset is a subgroup.
4. $\phi^{-1}(H)$ is a subgroup of G if H is a subgroup of G' given a homomorphism $\phi: G \rightarrow G'$.
5. Kernel and image of a homomorphism are subgroups.
6. Example: subgroup of S_3 , $\mathbb{Z}/n\mathbb{Z}$, $\mathbb{C}^\times$, $C^0([0, 1])$.
7. Subgroup generated by a subset. Finitely generated groups: there exists $F(\{a_1, \dots, a_n\}) \twoheadrightarrow H$.
8. Subgroup of $\mathbb{Z}$ is nice.
9. Injective is just kernel is trivial.
10. HW: read section 6. Do 6.4, 6.5, 6.6, 6.7, 6.8, 6.9, 6.12, 6.14, 6.15, 6.16.